Phase Diagrams: Quiz 1 with Solutions

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Scope and notation. This document restates the five problems of Quiz 1 of the phase-diagram course and presents a solution to each problem. All the diagrams were redrawn with TikZ. The diagrams of Problems 2 and 5 were reconstructed from the coordinates of the original figures, and the lens of Problem 2 was computed for ideal solid and liquid solutions. The composition X is the mole fraction of the second component B, and T is the temperature. The symbols L and G denote the liquid and the gas, and Greek letters denote solid phases. Each three-phase reaction is written in the direction of cooling. Section and table numbers that refer to the lecture notes are marked "of the notes".

1 Problems

Problem 1. The binary system A–B forms the congruently melting compound A_xB_y. Alloys between A and A_xB_y are of the eutectic type, and alloys between A_xB_y and B are of the peritectic type. Draw a phase diagram for this system. Describe how the percentages of the phases change if the eutectic point moves slightly to the left or to the right.

Problem 2. Figure 1 shows an isomorphous system. A bar of A and a bar of B are joined and heated to T_2 for isothermal diffusion. For the average compositions x_1 and x_2, state which phases form and give their compositions.

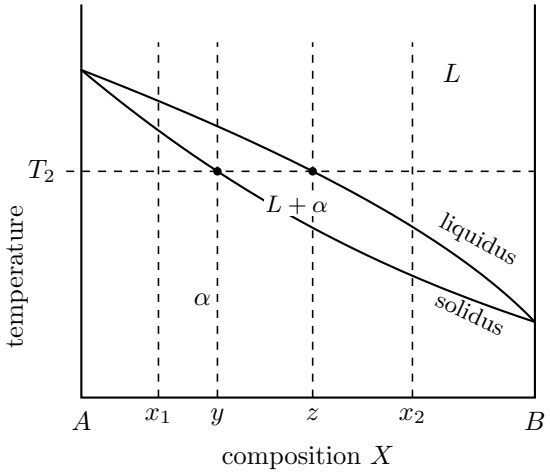


Figure 1: Isomorphous system of Problem 2, with the isotherm T_2 and the compositions x_1, y, z and x_2.

Problem 3. Figure 2 shows the P–T projection of a binary system with a quadruple point Q of the eutectoid type. Draw the T–X diagram at the pressure p_0.

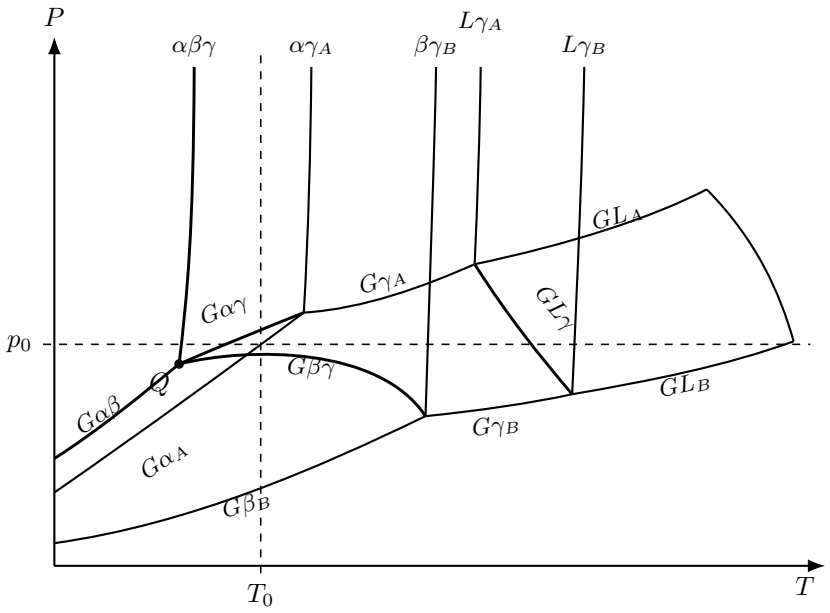


Figure 2: P–T projection of Problem 3. Thick lines: binary three-phase equilibria. Thin lines: equilibria of the pure components A and B. Dashed lines: the isobar p_0 and the isotherm T_0.

Problem 4. Figure 3 shows the Ag–Pr phase diagram. List all the three-phase reactions that involve the liquid phase.

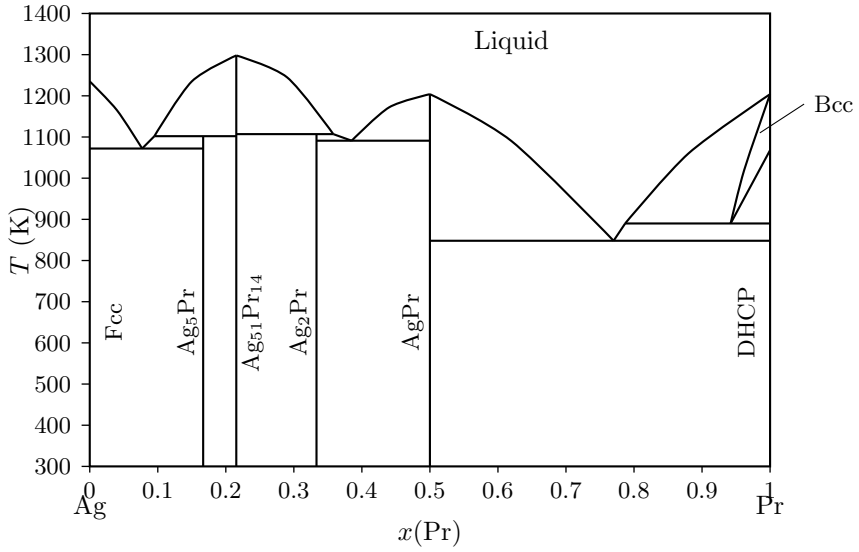


Figure 3: Ag–Pr phase diagram of Problem 4, redrawn from the quiz figure.

Problem 5. Figure 4 shows the diopside–anorthite–albite (Di–An–Ab) system. The Di–An and Di–Ab sides are binary eutectic systems, and the Ab–An side is a solid-solution loop without a minimum. Mark the eutectic points, the direction of falling temperature and the three-phase region, and state the meaning of each kind of line. Draw an isotherm through l_4. For solidification, give the approximate composition of point A and the path of the liquid during the cooling of composition A.

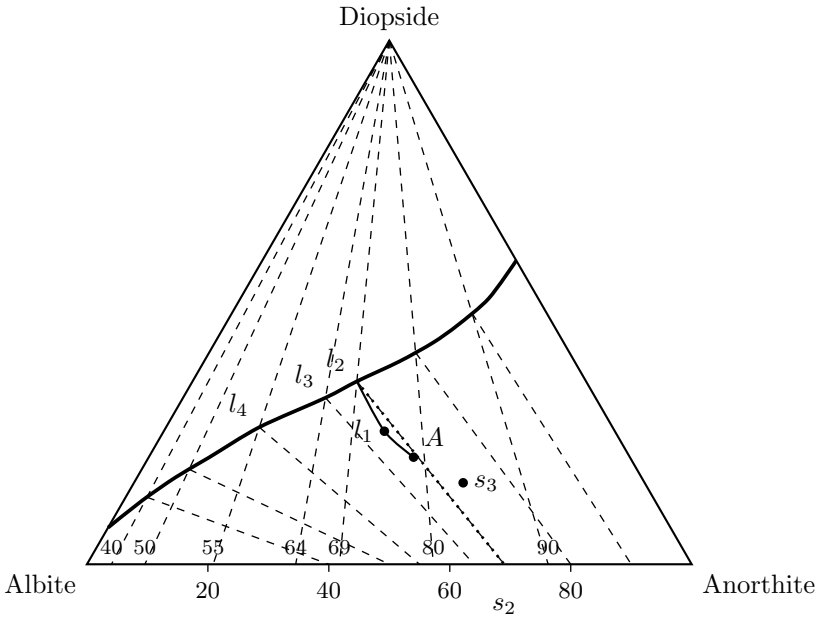


Figure 4: Di–An–Ab system of Problem 5 (compositions in wt%). Thick curve: cotectic. Dashed lines: diopside–liquid lines and liquid–plagioclase tie lines, with the An content of the plagioclase in equilibrium with each liquid. Solid curve and dotted line: crystallization of composition A.

2 Solutions

2.1 Problem 1: Compound with a Eutectic and a Peritectic Side

Construction from two partial diagrams. The compound A_xB_y, denoted γ, divides the system into the partial systems A–γ and γ–B. Figure 5 shows the diagram assembled from a eutectic diagram for A–γ and a peritectic diagram for γ–B. In the A–γ part, the liquid decomposes at the eutectic temperature T_E into the A-rich solid solution α and γ, $L \to \alpha + \gamma$. In the γ–B part, the B-rich solid solution β has the highest melting point, and the peritectic reaction $L + \beta \to \gamma$ occurs at T_P. The phase γ dissolves some B, so the γ field widens toward B and meets the peritectic isotherm.

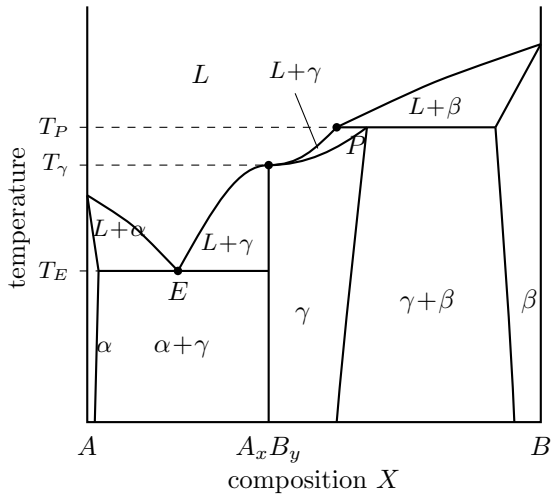


Figure 5: Phase diagram for Problem 1. E: eutectic point. P: liquid composition of the peritectic reaction. Dot at T_γ: congruent melting point of A_xB_y.

Congruent melting next to a peritectic. In a peritectic reaction, the product lies between the two reactants on the isotherm. For $L + \beta \to \gamma$, the liquid composition X_P therefore lies on the A side of the γ composition on the isotherm, between the stoichiometric composition X_γ and the γ boundary. A liquid can exist at X_P on the isotherm only if T_P exceeds the melting point T_γ of the compound. The liquidus therefore rises continuously from E through the congruent point (X_γ, T_γ) to P. The Gibbs–Konovalov relation (Section 7.6 of the notes) requires horizontal tangents where the liquid and the solid have the same composition. The congruent point of γ is therefore a horizontal inflection of the liquidus and not a maximum. A liquidus maximum at X_γ would make the γ liquidus fall toward B, and no peritectic reaction with only L, β and γ could then form on the B side. A convex-hull calculation with a compound that dissolves B but not A reproduced this topology.

Total amounts of the phases. Consider an alloy of composition X_0 between X_α and X_γ, the compositions of α and γ on the eutectic isotherm. Below T_E, the lever rule gives the fractions of the two phases as

$$n_\alpha = \frac{X_\gamma - X_0}{X_\gamma - X_\alpha}, \qquad n_\gamma = \frac{X_0 - X_\alpha}{X_\gamma - X_\alpha}, \tag{2.1}$$

where n_α is the fraction of α and n_γ is the fraction of γ. These fractions do not contain the eutectic composition X_E, so a small shift of the eutectic point leaves the total amounts of α and γ unchanged.

Amounts of primary crystals and eutectic. The shift changes the microconstituents, which are the primary crystals formed above T_E and the eutectic mixture formed at T_E. For a hypoeutectic alloy, $X_\alpha < X_0 < X_E$, the lever rule on the eutectic isotherm gives

$$n_{\text{eut}} = \frac{X_0 - X_\alpha}{X_E - X_\alpha}, \qquad n_{\text{prim}} = \frac{X_E - X_0}{X_E - X_\alpha}, \tag{2.2}$$

where n_{eut} is the fraction of eutectic and n_{prim} is the fraction of primary α. If the eutectic point moves to the right, X_E increases, so the alloy contains more primary α and less eutectic. The eutectic itself then contains more γ, with the fraction $(X_E - X_\alpha)/(X_\gamma - X_\alpha)$. A shift to the left has the opposite effect. If X_E passes X_0, the alloy changes from hypoeutectic to hypereutectic, and primary γ replaces primary α. An alloy of the original eutectic composition contains, after a small shift in either direction, a small amount of primary α or primary γ in addition to the eutectic.

2.2 Problem 2: Isothermal Diffusion in an Isomorphous System

Layers during diffusion. At T_2, pure A is solid, and pure B is liquid, because T_2 lies above the melting point of B (Fig. 1). At the joint, A dissolves into the liquid and B diffuses into the solid. At fixed temperature and pressure, both chemical potentials are constant across a two-phase mixture, so no diffusion flux passes through a two-phase zone. The $L + \alpha$ region of the diagram therefore appears only as a sharp interface. The couple consists of a solid α layer with compositions from 0 to y and a liquid layer with compositions from z to 1 (Fig. 6). The tie line at T_2 fixes the compositions y and z on the two sides of the interface.

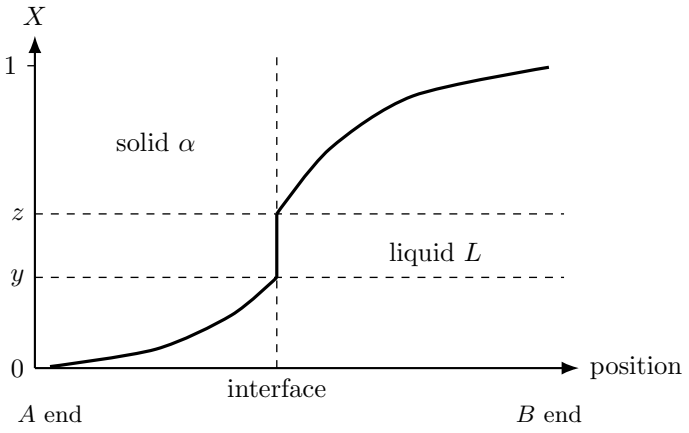


Figure 6: Composition profile of the diffusion couple at T_2 before the composition becomes uniform.

Final state. Diffusion continues until the composition is uniform. For the average composition $x_1 < y$, the final state lies in the α field, so the liquid disappears and a single solid α of composition

x_1 remains. For the average composition $x_2 > z$, the final state lies in the liquid field, so the solid dissolves completely and a single liquid of composition x_2 remains. Only an average composition between y and z leads to a final mixture of α with composition y and liquid with composition z, in the amounts given by the lever rule.

2.3 Problem 3: T–X Diagram from a P–T Projection

Crossings of the isobar. In Fig. 2, the isobar p_0 lies above Q and below the triple point of pure A at which α, γ and gas coexist. The isobar crosses seven curves, which are listed in Table 1 in order of increasing temperature. Each binary three-phase curve becomes a three-phase isotherm of the T–X diagram, and each curve of a pure component becomes a transition point on the side $X = 0$ or $X = 1$. Pure A forms neither γ nor liquid at p_0 and sublimes directly at T_3.

Table 1: Curves of Fig. 2 crossed by the isobar p_0.

Temperature	Curve	Equilibrium	Reaction on cooling
T_1	$\alpha\beta\gamma$	binary, three phases	$\gamma \to \alpha + \beta$ (eutectoid)
T_2	$G\alpha\gamma$	binary, three phases	$G \to \alpha + \gamma$
T_3	$G\alpha_A$	pure A	sublimation of α
T_4	$\beta\gamma_B$	pure B	$\gamma \to \beta$
T_5	$GL\gamma$	binary, three phases	$L \to G + \gamma$
T_6	$L\gamma_B$	pure B	solidification
T_7	GL_B	pure B	condensation

Type of each three-phase isotherm. At T_1, the γ solid solution decomposes on cooling into α and β, the eutectoid reaction that defines the type of Q. At T_2, gas forms below the sublimation temperature T_3 of pure A. The partial pressure of A in a gas that also contains B is smaller than p_0, so α coexists with this gas at a lower temperature than with pure A vapor. The gas is therefore the middle phase, and the reaction on cooling is $G \to \alpha + \gamma$. At T_5, the liquid is the middle phase, and the reaction on cooling is $L \to G + \gamma$, a eutectic-type reaction with a gas phase (Table 3 of the notes).

Assembly of the diagram. Figure 7 shows the resulting diagram. Below T_1, the alloys consist of α and β. Between T_1 and T_2, the γ field opens in the middle, with $\alpha + \gamma$ and $\gamma + \beta$ on the two sides. The isotherm at T_2 caps the $\alpha + \gamma$ region, and the $\alpha + G$ region rises from this isotherm to the sublimation point of A at T_3. The $\gamma + \beta$ region closes at the transformation point of B at T_4. The $G + \gamma$ region extends from T_2 to the isotherm at T_5, where the liquid appears. Above T_5, the $L + \gamma$ region closes at the melting point of B at T_6, and the $G + L$ region closes at the boiling

point of B at T_7. Two two-phase regions lie above each three-phase isotherm and one lies below, as required for eutectic-type reactions.

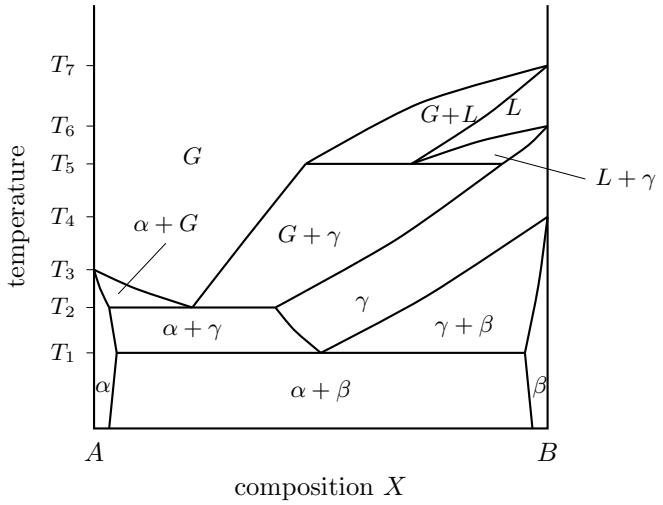


Figure 7: Schematic T–X diagram at the pressure p_0. T_1 to T_7: crossings of the isobar listed in Table 1.

2.4 Problem 4: Three-Phase Reactions in the Ag–Pr System

Identification of the reactions. Each horizontal line of Fig. 3 that touches the liquidus is a three-phase isotherm with liquid. The type of each reaction follows from the middle phase on the isotherm. A liquid in the middle gives a eutectic, a solid compound in the middle gives a peritectic, and a solid in the middle that decomposes into liquid and another solid on cooling gives a metatectic. The compositions and temperatures in Table 2 were read from the diagram, with an accuracy of ± 10 K and ± 0.01 in x(Pr).

Table 2: Three-phase reactions with liquid in the Ag–Pr system.

Reaction on cooling	Type	T (K)	x_L(Pr)
$L \rightarrow \mathrm{Fcc(Ag)} + \mathrm{Ag_5Pr}$	eutectic	1072	0.08
$L + \mathrm{Ag_{51}Pr_{14}} \rightarrow \mathrm{Ag_5Pr}$	peritectic	1102	0.10
$L + \mathrm{Ag_{51}Pr_{14}} \rightarrow \mathrm{Ag_2Pr}$	peritectic	1107	0.36
$L \rightarrow \mathrm{Ag_2Pr} + \mathrm{AgPr}$	eutectic	1091	0.39
$\mathrm{Bcc(Pr)} \rightarrow L + \mathrm{DHCP(Pr)}$	metatectic	890	0.79
$L \rightarrow \mathrm{AgPr} + \mathrm{DHCP(Pr)}$	eutectic	848	0.77

Two-phase equilibria with liquid. The compounds $Ag_{51}Pr_{14}$ and AgPr melt congruently, at 1298 K and 1204 K, and pure Ag melts at 1235 K and pure Pr at 1204 K. These equilibria involve only two phases and are therefore not three-phase reactions. The transformation of pure Pr from Bcc to DHCP at 1068 K involves no liquid.

2.5 Problem 5: The Diopside–Anorthite–Albite System

Eutectic points and falling temperature. The two binary eutectics lie on the sides of the triangle (Fig. 8). The eutectic E_1 on the Di–An side lies at 58 wt% Di and 42 wt% An, and the eutectic E_2 on the Di–Ab side lies close to the Ab corner, at 7 wt% Di. The Ab–An side has no minimum, so the system has no ternary eutectic. Temperature falls along the cotectic from E_1 to E_2, along the Ab–An side from An to Ab, and in each primary field from the Di corner or the An corner toward the cotectic.

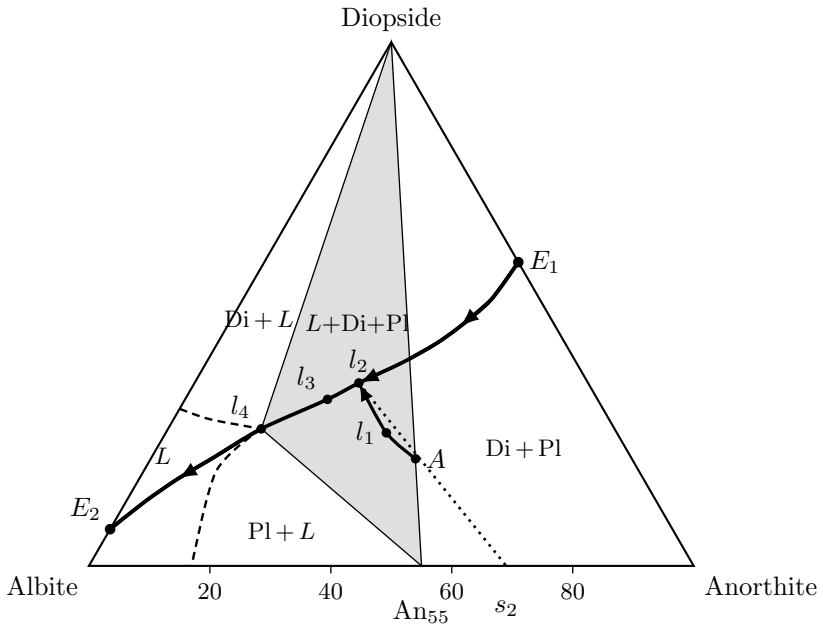


Figure 8: Solution of Problem 5. Arrows on the cotectic: direction of falling temperature. Shaded triangle: three-phase field at the temperature of l_4. Dashed curve through l_4: isotherm. Thick arrow: path of the liquid for composition A.

Meaning of the lines. The thick curve is the cotectic, along which the liquid is saturated with both diopside and plagioclase. The dashed lines from the Di corner are diopside–liquid tie lines, and their extensions to the Ab–An side mark the projection of each liquid from the Di corner. The dashed lines from the cotectic to the Ab–An side are liquid–plagioclase tie lines, and each

number gives the An content of the plagioclase in equilibrium with the liquid. The solid curve from A through l_1 to l_2 is the path of the liquid while plagioclase crystallizes alone. The dotted line through l_2, A and s_2 is the tie line at the moment the liquid reaches the cotectic, and the point s_3 is the bulk composition of the diopside and plagioclase crystallized when the liquid is at l_3.

Three-phase region and the isotherm through l_4. For each liquid on the cotectic, diopside, the liquid and plagioclase form a tie triangle, and these triangles together make up the three-phase region $L + \mathrm{Di} + \mathrm{Pl}$. At the temperature of l_4, the tie triangle connects Di, l_4 and plagioclase An_{55}, shaded in Fig. 8. The isotherm through l_4 has two branches, one in the diopside field to the Di–Ab side and one in the plagioclase field to the Ab–An side. The two branches form a V whose vertex at l_4 points toward higher temperature along the cotectic. At this temperature, the liquid field lies toward the Ab corner, the fields Di + L and Pl + L lie on the two sides of the triangle, and the field Di + Pl lies between the triangle and the An corner.

Composition A and the path of the liquid. Point A lies at 20 wt% Di, 44 wt% An and 36 wt% Ab, read from Fig. 4 with an accuracy of 2 wt%. Point A lies in the plagioclase field, so plagioclase crystallizes first. The plagioclase is richer in An than the liquid, and the liquid moves away from the plagioclase along the curved path from A through l_1 to l_2. The path curves because the plagioclase becomes richer in Ab as the temperature falls. At l_2, the liquid reaches the cotectic in equilibrium with plagioclase s_2 (An_{69}). Diopside then crystallizes together with plagioclase, and the liquid moves down the cotectic from l_2 through l_3 to l_4 while the plagioclase becomes more albitic. Equilibrium crystallization ends at l_4, where the join between Di and the plagioclase passes through A. The final plagioclase is An_{55}, the ratio An/(An + Ab) of composition A, and the solid contains 20 wt% diopside and 80 wt% plagioclase. For fractional crystallization, the liquid continues down the cotectic beyond l_4 toward E_2.